Buckthorn (and honeysuckle) sampling

Background information

Today, we are going out into the field to sample invasive species that are affecting woodlands around Western's campus. Below, you can see the area we will be sampling (from the university bridge to Western Alumni field). Our primary goals will be to (1) systematically sample the area for the presence of buckthorn and honeysuckle and (2) identify areas where large buckthorn trees are growing to prioritize their removal (haphazard sampling).



Health and Safety

We are going to be working off-trail in a woodland area near the river. Please watch your step, be careful of overhanging branches and thorns, and be mindful of uneven and slippery terrain. Some of the areas we will be working in will be damp and wet so make sure you have solid footing. Do not approach or go into the river, the fast-moving waters can be dangerous. Keep at least a 5 m buffer away from any moving water while you are sampling. If you are unable to safely sample one of your plots, please inform the instructor. Do not conduct any sampling in areas that may be unsafe. If you are injured, please let the instructor know immediately.

Phase I: Systematic sampling

Objective: Estimate buckthorn and honeysuckle densities and determine their correlation with environmental variables

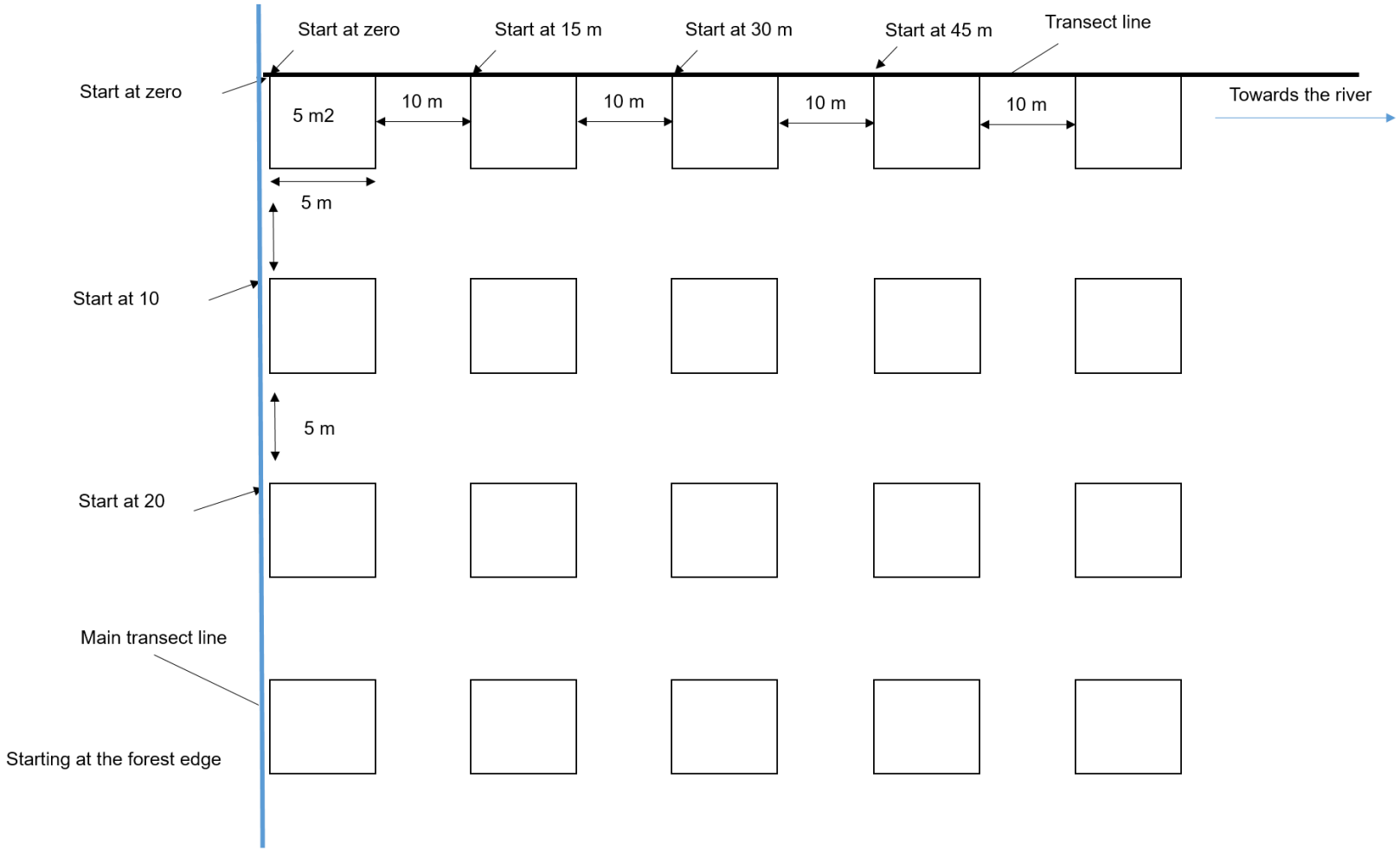
We are going to systematically sampling the wooded areas behind the soccer fields for both buckthorn and honeysuckle. These invasive species are out-competing native species in the area and taken over the riparian habitat around campus. The QR codes below link to documents that will help you identify both species in the field.

Buckthorn ID

Honeysuckle ID

Sampling design

Systematic sampling is a type of sampling technique that surveys regularly-spaced plots across environmental gradients. Below is the systematic sampling design we will be using today.



Measurements

In each of your plots you will be working to measure different characteristics of the environment and the invasive species you find. Below is a list the measurements you need to record.

Measure	Description
Geolocation (Latitude-longitude)	The easiest way to get your location is to use the GPS on your phone. Write down the GPS coordinates for the northwest corner of your $5m^2$ plots.
Distance from river	Use your transect tape to measure the distance to the river from your location (you can determine this after you are done measuring other variables).
Distance to closest trail	Determine how far away from the walking trail your plot is by measuring the distance from the centre of the plot.
Canopy cover	From the centre of your plot, look straight up and determine how much of the area is covered by the overlying canopy. For example, if you can see the sky in 60% of the area directly above your plot, your canopy cover would be 40%.
% Bare soil	How much of your plot is covered by bare soil?
% Herbaceous cover	How much of your plot is covered by herbaceous (non-woody) plants?
Species richness	Count the number of different plant species you see in each of the plots. You don't necessarily have to identify which species it is as long as you can tell that it is a different plant.
Buckthorn %	What percentage of the plot is covered in buckthorn?
Honeysuckle %	What percentage of the plot is covered in honeysuckle?
Stem circumference of buckthorn and honeysuckle	If buckthorn or honeysuckle is present in your plot, measure the largest

Phase II: Identifying high seed producers

Background information:

Buckthorn has an incredibly high rate of seed germination and the largest seed producers can substantially contribute to the expansion of an invasion. Therefore, identifying the largest producers for removal can have a disproportionate effect on invasive species spread.

Haphazard sampling is a form of sampling where we move along an area and search for our item of interest without any randomization or systematic plan. We will be moving through the wooded area together trying to identify any areas where there are large buckthorn tees, which we will classify as trees that are greater than 25 cm in circumference at the standard height of 1.4 metres. We will then be recording the GPS coordinates of each tree that reaches that threshold as we move along.

